

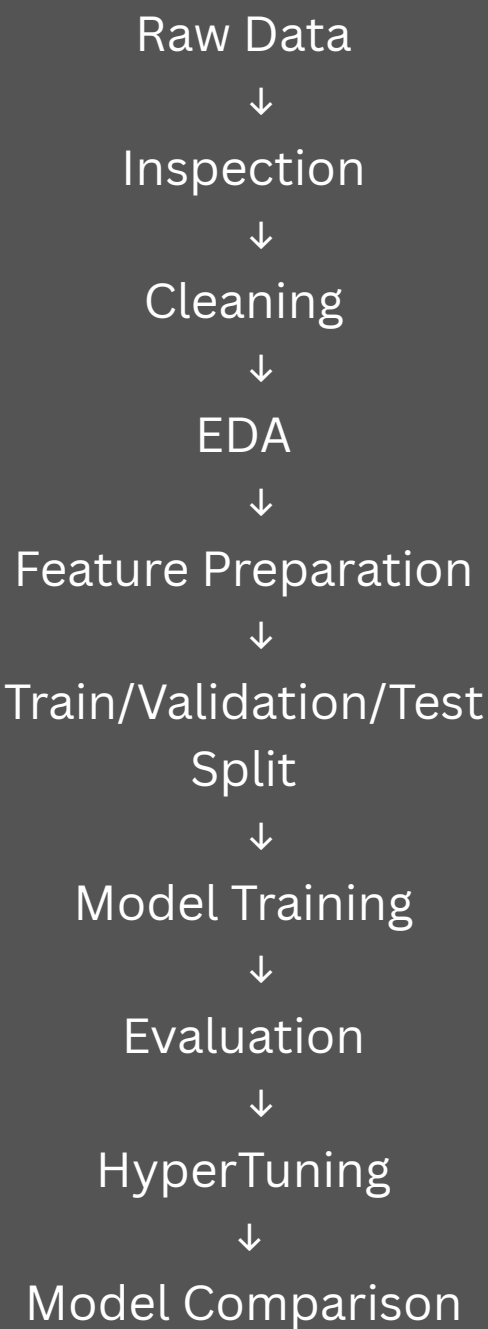
Day 6 -Pre-Dataset Readiness + Mentor Review

Progress Completed :

- ✓ Understood the UWOC problem statement and project objectives.
- ✓ Questions Discussed with MentorStudied the impact of turbidity (NTU) on optical signal transmission.
- ✓ Defined inputs, outputs, success metrics, and project scope.
- ✓ Designed the complete ML pipeline and experiment plan.
- ✓ Prepared dataset schema, risk checklist, and repository structure.

Mentor Discussion

1. Pipeline Discussion



2. Grid Search vs Random Search

Grid Search

- Tests every possible parameter combination.
- More exhaustive but computationally expensive.

Random Search

- Tests randomly selected parameter combinations.
- Faster and often finds good solutions with less computation.

3. Why RF Communication Does Not Work Well Underwater?

- Water strongly absorbs high-frequency radio waves.
- Signal attenuation is very high underwater.
- Communication range becomes extremely limited.

4. How Does Turbidity Affect Communication?

- Suspended particles absorb and scatter light.
- Received optical signal becomes weaker and noisier.
- Bit 0 and Bit 1 signal values may overlap.
- Fixed-threshold detection becomes unreliable.

Next Steps :

- Receive and inspect the dataset.
- Perform data quality audit.
- Apply approved preprocessing strategy.
- Begin baseline and ML model implementation.

